several differences in how they are used, including the following:

- Groups are used primarily to organize users, while roles are mainly used as a way to organize permissions.
- As discussed in the previous section, groups tend to be assigned centrally, whereas roles tend to be specific to a particular application or API. As an example, every API may have an `admin` role, but the set of users that are administrators may differ from API to API.
- Group-based systems often allow permissions to be assigned to individual users, but RBAC systems typically don't allow that. This restriction can dramatically simplify the process of reviewing who has access to what.
- RBAC systems split the definition and assigning of permissions to roles from the assignment of users to those roles. It is much less error-prone to assign a user to a role than to work out which permissions each role should have, so this is a useful separation of duties that improves security.

RBAC is almost always used as a form of mandatory access control, with roles being described and assigned by whoever controls the systems that are being accessed. It is much less common to allow users to assign roles to other users as they can with permissions in discretionary access control approaches. Instead, it is common to layer a DAC mechanism such as OAuth 2.0 (chapter 7) over an underlying RBAC system so that a user with a moderator role, for example, can delegate some part of their